DAY 10 OF 10

Mini Project & What Next

Build a complete program and plan your learning path

Today's agenda

1. Project: Student Marks Manager
2. Clean code habits
3. Git and GitHub in one slide
4. Interview questions
5. Your learning roadmap

Python in 10 Days
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Project: Student Marks Manager

- A command line app to add students, view all, find the topper
- Saves data in a JSON file so nothing is lost when the program closes
- Uses everything from this course:
  - functions, dictionaries, loops, conditions
  - file handling with JSON, exception handling
- Menu: 1 Add 2 Show 3 Topper 4 Quit

Project code: part 1 (data helpers)

- load() reads the JSON file; if it does not exist yet it returns an empty dict
- save() writes the dictionary to disk
- add_student() reads input, validates the marks and saves

Python

```python
import json

FILE = "students.json"

def load():
    try:
        with open(FILE) as f:
            return json.load(f)
    except FileNotFoundError:
        return {}

def save(data):
    with open(FILE, "w") as f:
        json.dump(data, f, indent=2)

def add_student(data):
    name = input("Name: ").strip().title()
    try:
        marks = float(input("Marks (0-100): "))
    except ValueError:
        print("Marks must be a number")
        return
    if not 0 <= marks <= 100:
        print("Marks must be between 0 and 100")
        return
    data[name] = marks
    save(data)
    print("Saved!")
```

Project code: part 2 (menu loop)

- show_all() prints a neat table with f-string alignment
- topper() uses max(data, key=data.get)
- main() runs the menu in a while True loop until the user quits

Python

```python
def show_all(data):
    for name, marks in sorted(data.items()):
        print(f"{name:<15}{marks:>6.1f}")

def topper(data):
    if data:
        name = max(data, key=data.get)
        print("Topper:", name, data[name])
    else:
        print("No students yet")

def main():
    data = load()
    while True:
        print("\n1.Add  2.Show  3.Topper  4.Quit")
        choice = input("Choose: ")
        if choice == "1":
            add_student(data)
        elif choice == "2":
            show_all(data)
        elif choice == "3":
            topper(data)
        elif choice == "4":
            break
        else:
            print("Invalid choice")

main()
```

Level up the project (ideas)

- Add delete and update options
- Show class average and a pass / fail count
- Store subject-wise marks using nested dictionaries
- Convert the functions into a StudentManager class (Day 8)
- Load the JSON into Pandas and draw a bar chart (Day 9)
- Share it on GitHub with a README: this is portfolio material

Clean code habits

DAY 10

- Follow PEP 8: 4 spaces, snake_case names, lines under about 88 characters
- Meaningful names beat comments: total_price, not tp
- Small functions that do one job
- Handle errors where input comes from outside (user, files, network)
- Use virtual environments and a requirements.txt
- Read errors from the bottom line upward: that is where the cause is

Git and GitHub in one slide

- Git tracks versions of your code; GitHub hosts it online
- Employers look at your GitHub, so push your practice and projects

Python

```
git init
git add .
git commit -m "Student marks manager"
git branch -M main
git remote add origin <your-repo-url>
git push -u origin main
```

Common interview questions

- Difference between list and tuple? (mutable vs immutable)
- What is a dictionary and how does it differ from a set?
- What are *args and **kwargs?
- Explain mutable vs immutable types with examples
- What is a decorator / generator? (study next)
- Shallow copy vs deep copy; == vs is
- How do you handle exceptions? What is finally?
- What is a list comprehension? Write one

Your learning roadmap after Python

DAY 10

Step	Topic	What you will learn
1	SQL	Queries, joins, aggregations, window functions
2	Data analysis	Pandas deep dive, visualisation, EDA, Excel / Power BI
3	Machine Learning	Regression, classification, model evaluation
4	Deep Learning	Neural networks, CNNs, TensorFlow / PyTorch
5	Generative AI	LLMs, prompt engineering, RAG, agents
6	Projects + GitHub	3 to 5 solid projects for your resume

Day 10 recap & what to do next

DAY 10

- You can now read and write real Python programs
- Keep practising daily: 30 to 45 minutes beats a 5-hour weekend binge
- Build one small project every week
- Practice sites: HackerRank, LeetCode (easy), Kaggle (datasets)
- Stay tuned for the SQL and Machine Learning batches
- Thank you, and keep coding!